Project Submission Cover Sheet

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Implications of AI and AI Adoption in Real Estate for BlackRock

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Executive Summary

The Global Consulting Project was conducted for BlackRock's Global Real Estate Research team. It is aimed to explore the transformative role of Artificial Intelligence in shaping the future of real estate investments. As the world's largest asset manager, BlackRock is uniquely positioned to lead on emerging technologies but with the rise of generative AI and machine learning comes both opportunity and uncertainty. The central question guiding our project was: "How can BlackRock responsibly and effectively adopt AI to enhance its real estate investment capabilities and decision-making processes?"

The scope of the project covered three primary areas: (1) How AI is influencing real estate demand and human behaviour, (2) The application of AI within real estate investment strategy and (3) The risks and limitations associated with AI adoption. To answer these, we combined qualitative primary research, conducting interviews with AI experts, academics and real estate professionals, with extensive secondary research that included academic publications, industry reports and real-world case studies. We also integrated feedback from our client and academic supervisor throughout the process.

Our findings reveal that AI is acting as a fourth major wave of disruption in real estate, following mechanisation, urbanisation and globalisation. It is altering demand patterns across asset classes, enabling a shift from traditional valuation drivers to real-time, intelligence-driven metrics. Data centres, logistics hubs and life sciences facilities emerge as clear beneficiaries of AI's infrastructure needs, while traditional retail and office spaces face headwinds unless repositioned with tech-enhanced services.

We identified a wide range of AI applications already delivering value across the real estate investment lifecycle. These include AI-powered tools for lease abstraction, deal underwriting, sentiment-based market analysis and advanced risk forecasting using machine learning and geospatial data. AI also supports operational efficiency through GPT-powered document analysis and internal tools like JLL GPT. A key innovation we proposed is the Structured AI Judgment Process, a framework combining AI outputs with

quantitative validation and feedback loops to ensure transparency, explainability and strategic accountability.

However, AI adoption also introduces challenges. These include data bias, regulatory compliance, lack of explainability and risks of overreliance on opaque models. To address these, we proposed a multi-layered mitigation strategy, including prompt engineering best practices, stop lists, classifier models and human-in-the-loop oversight. We highlighted Morgan Stanley's Private ChatGPT system as a benchmark example of responsible, institution-aligned AI deployment.

To guide BlackRock's strategic approach, we applied the Three-Horizon Framework, outlining a roadmap across short, medium and long-term opportunities. In Horizon 1 (0–1 year), we recommend enhancing data infrastructure and piloting proven AI tools. In Horizon 2 (2–5 years), we advise investing in scalable AI platforms, strategic partnerships and transparency frameworks. Horizon 3 (5+ years) focuses on embedding AI in strategic decision-making, portfolio rebalancing and investing in AI-era infrastructure such as digital twins.

This report concludes that while AI presents significant complexity, it also offers BlackRock a powerful opportunity to lead the industry in responsible, data-driven innovation. With the right safeguards and governance, AI can be a critical enabler of competitive advantage in real estate investing.

I. Aims

This project aimed to explore the transformative role of AI in shaping the future of real estate investment and strategy for BlackRock. The focus lied in three core areas:

1. *Understanding AI's Influence on Real Estate Demand and Human Behaviour*

To examine how AI-driven changes in consumer behaviour, lifestyle and business operations impact property valuations, location preferences and demand across various real estate classes.

2. *Integrating AI into Investment Strategy*

To identify how AI technologies can be effectively applied to enhance the overall real estate investment.

3. *Evaluating Risks and Challenges of AI Adoption*

To assess the key risks, regulatory and ethical concerns associated with AI integration in investment models and to develop practical mitigation strategies through a structured risk register, ensuring responsible and sustainable adoption.

By conducting research across these three areas, we ultimately aimed to provide the client with strategic recommendations on the future disruptive impact of AI on real estate demand, effective ways to leverage AI and limitations of the risks associated with AI adoption.

II. Context

This project was commissioned by BlackRock Inc., the world's largest asset manager with nearly $12 trillion in assets under management as of December 2024. Against a backdrop of rapid technological advancement, increasing public interest in AI and growing regulatory scrutiny, BlackRock sought to explore how AI could be a source of competitive advantage for their global portfolio of direct investments in commercial real estate.

The team at BlackRock who engaged us was the Real Estate Research Team, who actively assist investment teams across multiple geographies and asset classes. This team is especially focused on understanding long-term trends, particularly any trend which may impact real estate demand or portfolio stability. AI emerged as a priority topic for them not only because of its transformative potential but also due to its opacity and the emerging need for clear frameworks to evaluate its risks and opportunities.

From a broader industry perspective, AI is already beginning to alter the structure of real estate markets, including how tenants interact with the built environment and how

investors identify and underwrite deals. For institutional investors such as BlackRock, this presents a need to capture the upside of AI-enhanced tools while navigating risks such as black-box algorithms, data privacy concerns and regulatory uncertainty.

III. Methodology

To answer key research questions set out for us, we used a combination of research approaches. This included qualitative primary research through expert interviews and quantitative & qualitative secondary research from academic and industry sources. This dual approach ensured both depth and breadth of research to cover the entire scope of all the key research questions.

The team started by focusing on primary research, as we know that securing these interviews would be time-consuming and critical to providing unique insights for the client. We focused on three types of experts: academic researchers, industry professionals and AI-specific experts. This approach was adopted to ensure we gained a holistic understanding of how AI is currently utilised across the real estate industry and to explore the future implications for each asset class. The team utilised the Cambridge academic network, our own personal networks as well as the client network to achieve this. This process resulted in over thirty reach-outs, more than ten informal conversations and five formal research interviews. Two of the formal interviews were conducted with academic researchers, two with industry experts and one with an AI-specific expert, which accomplished our goal for a well-rounded group of interviews. All interviews were conducted using a questionnaire agreed upon by the team, with detailed notes taken and team debriefs afterwards to ensure that all information gathered was disseminated and analysed by all members of the team.

Given BlackRock's extensive research capabilities and discussions had with them early in the process, we understood that secondary research would be less likely than primary research to provide the unique insights that they were looking for as part of the deliverables. As such, secondary research was undertaken primarily to broaden our understanding of trends, validate and support information found in primary research and

find case studies on AI adoption which may be of interest to our client. To conduct efficient secondary research, topics were broken down based on the key research questions and assigned to individual team members, with collaborative reviews happening every other day to share findings amongst the team.

As information came to light via this process, the team applied the most insightful findings to each focus area of the PID. Critically, during the collaborative research reviews, the team also introduced original intellectual contributions and our own viewpoints on findings to exceed the client's expectations. Further improving the quality and depth of thinking were weekly meetings with the client, where we followed a structured agenda to review key findings and discuss topics of interest and weekly meetings with our academic supervisor where we talked through our research methods and uncovered new sources.

IV. Analysis

The process to move from research to concrete findings, results and recommendations included a mixed-methods approach, combining qualitative expert interviews with secondary data analysis and industry case studies.

The first PID objective, how AI would influence real estate demand and human behaviour, was addressed by examining the results of perspectives from the primary research interviews, learning from past trends and corroborating them with macro-level trends obtained from secondary sources. The team focused on the likely impacts of these findings on to each major real estate asset class to create a matrix categorising asset class as "front runners", "marginal gainers" or "stragglers".

Analysing the second objective, the potential for AI integration in BlackRock's investment strategy, required a more granular understanding of where AI could be used across a real estate investment workflow. The team focused on areas where AI integration could yield improvements in decision making accuracy or efficiency. Various use-cases were reviewed and their relevancy to BlackRock's process and strategy was contemplated.

The final objective, identifying risks and challenges with AI adoption, required the development of a risk matrix to cover operational, financial, reputational, workforce and legal risks. As risks were identified through research they were logged in the matrix and categorised and low, medium or high risk. Accompanying mitigations were included if pre-existing or hypothesised by the team if not.

From the risk analysis the concept of a structured AI judgement process was born. We found that one of the largest risks of AI adoption was that results came from a "black box" and were generally unexplainable. This posed a large issue for a company such as BlackRock who must disclose and explain decisions to investors, clients and the public. Therefore, we developed a model with a three-stage framework: data augmentation, hybrid AI-quantitative model integration and feedback loops. The underlying rationale was that AI outputs must be not only accurate but also explainable and "priceable", a distinction reflecting the importance of investor confidence in "black box" decision making.

The analytical process consistently aligned with the PID's stated objectives, bridging the gap from exploration to making recommendations. By blending qualitative insight with real-world use-cases, the team delivered a set of strategic tools that were both innovative and applicable to our client.

V. Results

Understanding AI's Influence on Real Estate Demand and Human Behaviour

Our research led us to conclude that AI represents the fourth major wave of disruption in real estate, following mechanisation, urbanisation, and globalisation. Unlike previous transformations, AI doesn't just change where and how people live and work but fundamentally reshapes the value drivers of real estate itself. As AI becomes increasingly intelligent and cost-effective, driven by exponential advancements and significant step-changes in computing power (Slumbers, 2025), real-time intelligence, automation, and predictive analytics will continue to transform how assets are valued, operated, and exited.

This shift is producing winners and losers among asset classes (Anderson, 2025). Data centres are clear front-runners, propelled by surging demand for digital infrastructure to support AI models. Logistics assets also benefit from AI-optimised supply chains and the expansion of e-commerce. Similarly, life sciences and healthcare spaces are in demand due to their alignment with innovation economies and university clusters.

In contrast, traditional office and retail assets face growing structural risk. AI-enabled remote work and virtual collaboration are reducing the need for physical office space. Similarly, AI-driven e-commerce is accelerating the decline of brick-and-mortar retail. Without strategic repositioning, these assets risk obsolescence (Slumbers, 2025). For BlackRock, the implication is clear: portfolio exposure must be adjusted toward tech-aligned, service-oriented and future-resilient assets.

More broadly, Lindenthal (2025) believed that real estate is becoming increasingly service centric. AI enables buildings to function as dynamic, responsive environments rather than static structures. Landlords are expected to behave more like service providers, delivering personalisation, operational efficiency and real-time feedback. For investors, asset management capabilities are now just as important as acquisition decisions.

AI Applications: Enhancing the Investment Lifecycle

AI is delivering real, measurable benefits across the real estate investment lifecycle, from deal sourcing, underwriting to asset management and exit planning. Its core value lies in improving data-driven decision-making, operational efficiency and the ability to uncover early market signals.

In underwriting and due diligence, massive efficiency gains are made through a multitude of tools. One example highlighted was AI-powered lease abstraction though a case study where a US-based real estate firm used AI to review hundreds of tenant leases in only a few days during due-diligence, identifying over missing documents and supporting more accurate cash flow projections (Prophia, 2024). Another example shown was GPT-

powered tools to enhance productivity by turning plain text into Excel formulas and summarising complex documents, making early-stage deal underwriting faster and more accessible (GPTExcel, n.d.). Market analysis is also evolving with AI-enabled sentiment tools. Redfin, for instance, developed a system that scans reviews, news and social media to detect neighbourhood trends before they appear in transaction data, giving investors an informational edge (Hintz, 2022).

One of the most powerful applications of AI is its transformative impact on risk analysis. Advanced machine learning and natural language processing are being used to identify, quantify and simulate diverse risk types, ranging from market volatility and property depreciation to environmental and legal risks. For example, predictive models can run thousands of simulations to estimate downside exposure. Geospatial AI highlights climate risks like flooding or wildfire exposure, while computer vision tools such as ZestyAI's "Roof Age" assess physical asset risk via satellite images (ZestyAI, n.d.).

Across all these cases, high-quality, integrated data is foundational. Platforms like CoStar and CoreLogic centralise vast property datasets, supporting better forecasting and valuation. CoreLogic's AI-powered automatic valuation models, for example, enhance valuation confidence by allowing users to explore the drivers behind each estimate (CoreLogic, 2022). Firms are also embedding AI directly into internal operations. JLL GPT, a proprietary language model, empowers brokers with instant insights and speeds up document analysis, boosting both efficiency and client service (JLL, 2024). Together, these applications show how AI, when paired with high-quality data and explainability, can enhance both investment speed and confidence.

Evaluating Risks and Challenges of AI Adoption

While AI offers substantial benefits to real estate investment, it also introduces critical risks and limitations, particularly around transparency, bias and responsible use. For institutional investors like BlackRock, ensuring that AI tools are trustworthy, explainable and compliant with regulations is essential.

A central challenge is explainability. Many AI systems, especially large language models, operate as black boxes, making it difficult to justify investment recommendations. This undermines both internal confidence and client trust. Tools like Shapley Additive Explanations (SHAP) and Local Interpretable Model-Agnostic Explanations (LIME) help address this issue. SHAP visualises which features most influenced a model's decision, while LIME offers plain-language explanations for specific outputs, such as why one property was preferred over another (Keita, 2023). A more advanced application is the structured AI judgment process, which integrates AI outputs with quantitative validation to ensure assumptions (like IRR or utility costs) are transparent and testable.

Another key limitation is biases in training data. Real estate models often rely on datasets that may be incomplete or historically skewed, such as overrepresentation of major cities or legacy pricing patterns (Boeing, 2019). To mitigate this, firms must audit data sources, diversify training inputs and use synthetic data to test for edge cases and underrepresented segments. Data need to be consistent and thorough (Lindenthal, 2025).

Compliance and ethics are also major concerns, particularly in real estate, where regulations like fair housing laws prohibit discriminatory practices. By designing prompts that explicitly prohibit bias, using stop lists to block high-risk terms and deploying classifier models to screen AI outputs, firms can prevent unintended violations while preserving model flexibility (National Institute of Standards and Technology, 2022).

Strong data governance and diversity are foundational, along with prioritising model explainability through transparent algorithms and summarisation prompts. These steps help ensure AI systems remain both compliant and trusted. A standout example is Morgan Stanley's custom ChatGPT, which used internal training data, built-in explainability and human reinforcement learning to align AI recommendations with institutional standards, proving responsible AI adoption is achievable at scale (Morgan Stanley, 2023).

Ultimately, these practices are not just safeguards, they position BlackRock to adopt AI confidently, ethically and competitively in a rapidly evolving investment landscape.

VI. Recommendations

We analysed BlackRock's key competitive advantages and in response to the inherent uncertainty surrounding AI implication and adoption applied the Three-Horizon Framework (Baghai, Coley and White, 1999, p. 45) to strategically manage their allocation of resources and attention across current, emerging and future opportunities.

In Horizon 1 (Immediate Term, 0-1 year), the recommendation emphasised investing in the foundational aspects of AI, primarily enhancing data infrastructure by consolidating and cleaning real estate data into a unified platform, akin to a "CoStar-like" database. Additionally, it suggests launching pilot projects in AI applications that have demonstrated maturity in the market. This includes integrating AI algorithms into deal analysis for rapid market data and property valuation verification, automating due diligence tasks through proven AI software such as reading AI for leases, contracts and financial statements and employing AI-driven scenario analysis for risk detection and anomaly identification. These pilot initiatives are complemented by productivity tools equipped with natural language query interfaces and interpretability tools like SHAP and LIME to ensure transparent AI-driven predictions (Ribeiro et al., 2016).

Horizon 2 (Medium Term, 2-5 years) centres on enhancing the foundational AI infrastructure, fostering a collaborative ecosystem through partnerships and closely collaborating with clients to co-develop AI applications. This involves further scaling the infrastructure by leveraging cloud computing, implementing robust data engineering and governance practices, and forming strategic partnerships with leading cloud and AI providers such as OpenAI. These collaborations can enable seamless integration, ensure model validation, and mitigate the risks associated with developing in-house models that may quickly become obsolete (Szumilo, 2025). Developing talent through targeted training programs will ensure the workforce is adequately equipped to leverage AI effectively.

Additionally, enhancing AI transparency by adopting the structural intelligence framework will be critical. Subsequently, AI applications will be extended into more advanced use cases such as scalable AI-driven portfolio management, real-time allocation optimisation and dynamic portfolio rebalancing based on real-time data feeds.

In Horizon 3 (Long Term, 5+ years), we anticipated AI-driven strategic transformation and new opportunities. Investments will shift even more towards AI-centric digital infrastructure such as data centres, cloud facilities and telecom sites. BlackRock should position itself strategically to meet rising demands in life sciences labs and technology incubator spaces. The long-term vision includes embedding AI deeply into strategic and investment decision-making processes, utilising advanced AI forecasting with synthetic data, perpetual digital twins and personalised investor updates through agentic generative AI (Romero-Cano, 2025). Lastly, aligning BlackRock's investments with emerging urban infrastructure needs covering digital, transportation and energy sectors, will ensure portfolio relevance and sustainable growth in the AI era.

Through this strategic roadmap, BlackRock will progressively build robust capabilities, capitalise on immediate opportunities and strategically position itself for long-term growth driven by advanced AI integration.

VII. Challenges

A key challenge of this project was the broad scope. We were tasked with examining the impact of artificial intelligence on society and linking these changes to trends in real estate investment. At the same time, we assessed how artificial intelligence is currently applied in investment strategy, identified associated risks and proposed ways to address them. The work involved both large-scale analysis, such as shifts in human behaviour and the influence of artificial intelligence on economic structures, and detailed strategic advice for a global asset manager. The PID also included complex questions on asset management, such as how artificial intelligence might enhance interactions with the built environment to improve operational efficiency. However, as the project progressed, the client advised us to concentrate on the investment-focused applications of artificial

intelligence. As a result, operational aspects were addressed only briefly in the appendix of the presentation sent to the client and excluded from the results mentioned above.

Another major challenge was conducting interviews with a wide variety of stakeholders. Given the overlap of artificial intelligence and real estate investment, it was essential to engage with individuals from a range of fields, including institutional investors, artificial intelligence experts, and academic researchers. Scheduling these discussions proved difficult, with time zone differences, limited availability and the need to tailor each conversation to the interviewee's expertise. It took significant effort to identify suitable contacts, craft personalised outreach messages and align interviews with our developing research structure. Consequently, these constraints resulted in a relatively low number of interviews, limiting the breadth of insights we could directly gather.

The cross-disciplinary nature of the topic also required us to interpret unfamiliar technical terminology and communicate it in a way that was relevant to business. Concepts such as model explainability, synthetic data and prompt engineering needed to be understood in enough depth to explain clearly to real estate professionals. At the same time, our recommendations had to be practical and aligned with the operational context of a company like BlackRock. Balancing technical insight with strategic relevance was challenging but ultimately rewarding.

To address these issues, our team adopted a proactive and cooperative approach. From the beginning, we maintained close communication with the client, which helped guide and shape our research. In terms of interviews, we made use of all available networks, including personal contacts, our academic supervisors and introductions from the client. This enabled us to reach qualified individuals through artificial intelligence, investment and academia. We created a standard interview guide to ensure consistency across conversations. Throughout the process, we also integrated feedback from our supervisor, particularly when refining technical content into useful business advice. This ongoing feedback loop allowed us to improve our narrative and deliver insights that were both robust and practical.

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